Laboratory Experiment No. 2

Objective: To construct and verify the functional operation of an 8:1 Multiplexer (MUX) and an 8-to-3 Priority Encoder using fundamental logic gate components.

Theoretical Background

1. 8:1 Multiplexer — Gate-Level Design

A multiplexer (MUX) is a data selector circuit that routes one of several input signals to a common output based on control signals. It functions essentially as a digitally controlled switch.

The 8:1 MUX configuration consists of the following lines:

- **Input lines:** D_0 through D_7 (eight data inputs)
- **Select lines:** S_2, S_1, S_0 (three control inputs)
- **Output:** Y (single output line)

The three select lines jointly determine which one of the eight data inputs is routed to the output Y at any given time.

Output Boolean Expression:

$$Y = D_0 \cdot S_2' \cdot S_1' \cdot S_0' + D_1 \cdot S_2' \cdot S_1' \cdot S_0 + D_2 \cdot S_2' \cdot S_1 \cdot S_0' + D_3 \cdot S_2' \cdot S_1 \cdot S_0$$
$$+ D_4 \cdot S_2 \cdot S_1' \cdot S_0' + D_5 \cdot S_2 \cdot S_1' \cdot S_0 + D_6 \cdot S_2 \cdot S_1 \cdot S_0' + D_7 \cdot S_2 \cdot S_1 \cdot S_0$$

Gate-Level Implementation Components:

- **AND gates** — used to generate individual minterms for each data input
- **NOT gates** — required to invert the select line signals
- **OR gate** — combines all individual minterm outputs to produce the final result

2. 8-to-3 Encoder — Gate-Level Design

An encoder performs the inverse operation of a decoder — it takes multiple input lines and produces a compressed binary code on fewer output lines. It is a many-to-one mapping circuit.

The 8-to-3 encoder specifications are:

- **Input lines:** I_0 through I_7 (eight active-high inputs)
- **Output lines:** A_2, A_1, A_0 (three-bit binary code output)

The circuit assumes that only one input is asserted HIGH at any given instant. This is referred to as a simple (non-priority) encoder.

Output Logic Equations (assuming one active input at a time):

$$\mathbf{A_2 = I_4 + I_5 + I_6 + I_7}$$

$$\mathbf{A_1 = I_2 + I_3 + I_6 + I_7}$$

$$\mathbf{A_0 = I_1 + I_3 + I_5 + I_7}$$

Each output bit is implemented using a single multi-input OR gate, making this circuit one of the simplest combinational encoders to realize in hardware.

Logisim Circuit Implementations

The circuits were simulated and verified using the Logisim digital logic simulator:

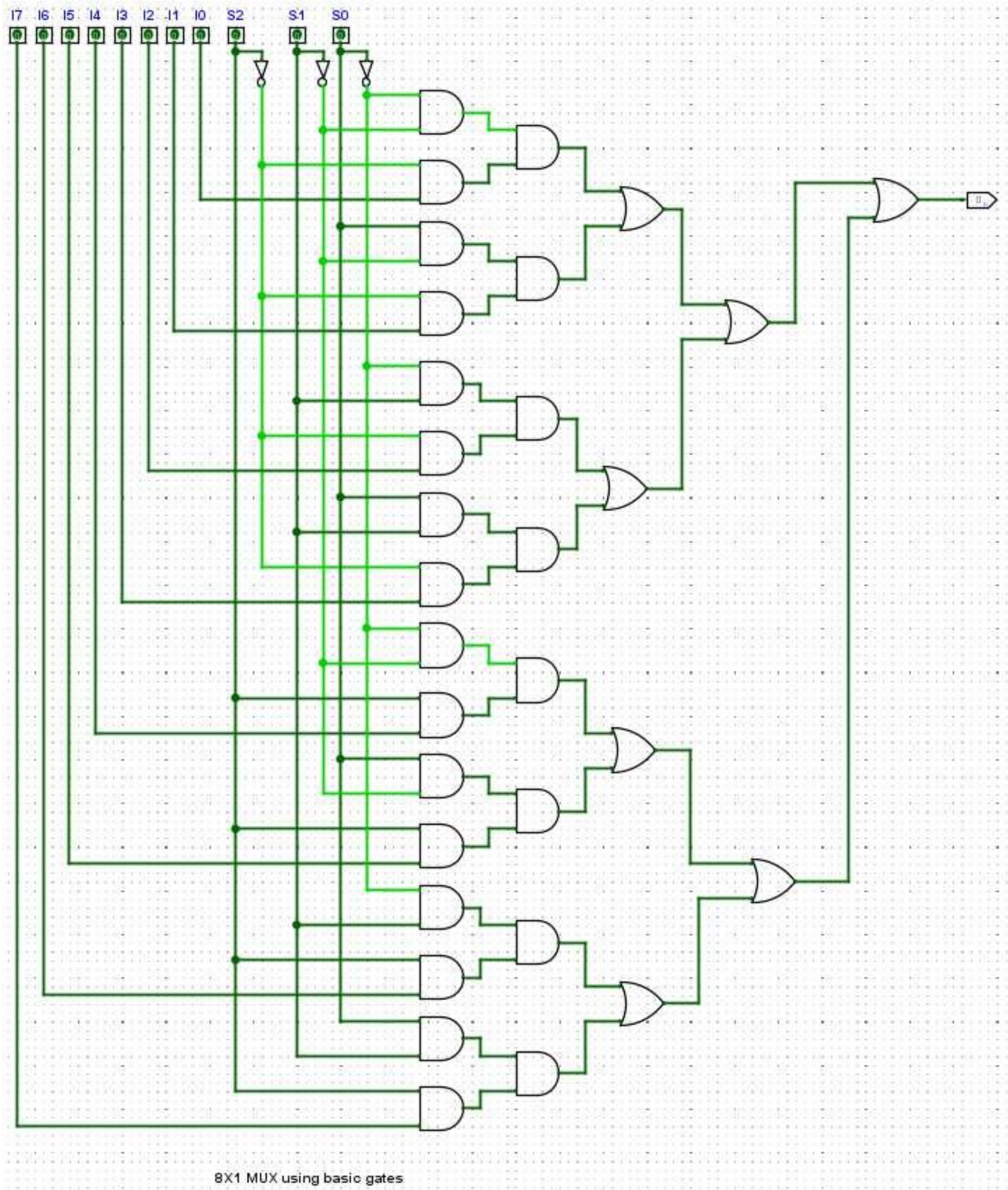

Figure 1: 8×1 Multiplexer Implemented Using Basic Logic Gates in Logisim

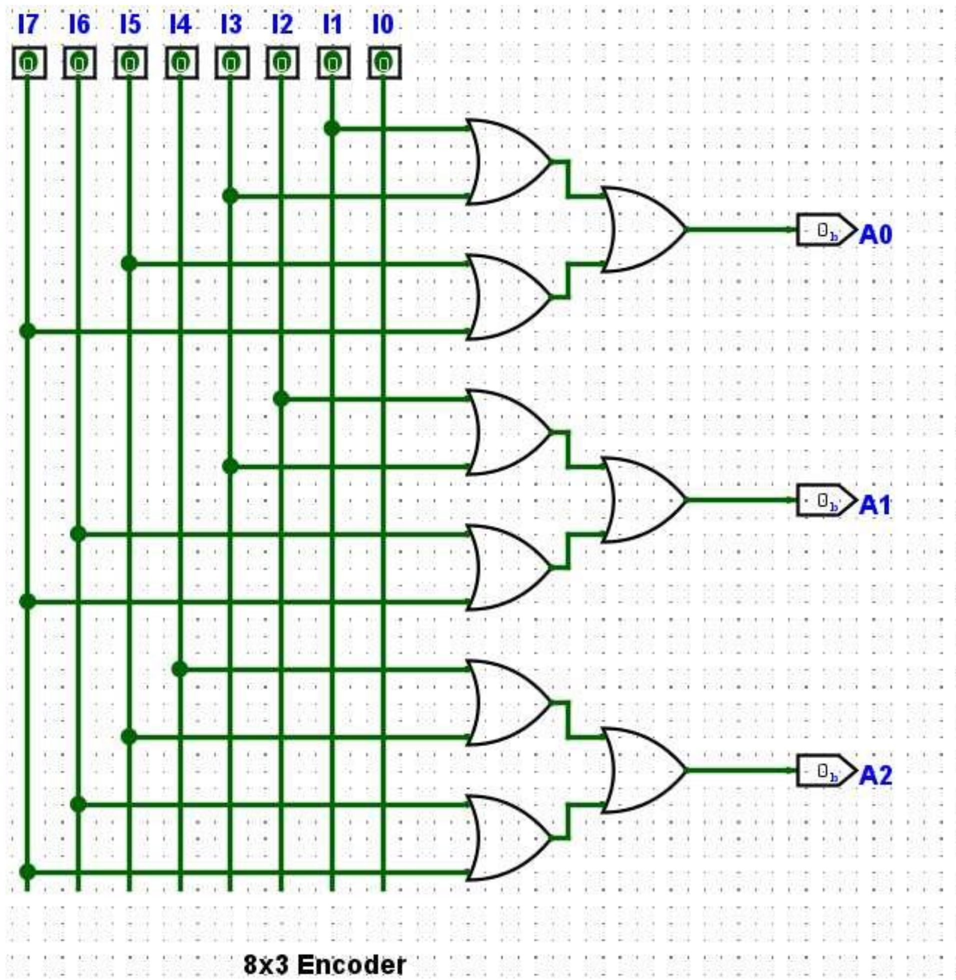

Figure 2: 8×3 Encoder Implemented Using OR Gates in Logisim